SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.E-CSE-AI	**Date:**	

Code of Conduct

I ____________Shaik Nuha(192472242)_______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____Sk.Nuha__________________

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:
Incorrect interpretation of ambiguous sentences
Failure to handle subject–verb agreement
Inefficient parsing for long queries
Example user query:
“Show me the transactions with the card from last month”
a) Analyze the ambiguity present in the given sentence using CFG
b) Evaluate the limitations of CFG in handling real-world conversational queries
c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
d) Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:
Backtracking issues
Poor handling of incomplete or ambiguous input
Difficulty in real-time response
Example command:
“Book a flight to Delhi with a window seat”
a) Analyze how the given sentence can lead to multiple parse structures
b) Analyze the limitations of top-down parsing in this real-time application
c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs
d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.
The system must handle:
Complex and lengthy medical sentences
Ambiguity in sentence structure
Accurate subject-verb agreement
Relationships between medical actions and entities (sub-categorization)
Real-time processing for hospital use
Example input:
"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."
The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.
a) Design a complete NLP architecture integrating:
CFG for syntactic structure
Probabilistic CFG (PCFG) for ambiguity resolution
Feature Structures for agreement handling
Efficient parsing techniques
b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).
c) Create a method to:
Incorporate sub-categorization frames for medical verbs
Handle real-time data processing
Improve accuracy and scalability in a hospital environment

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. CFG-Based Banking Customer Support Chatbot

Given Query

"Show me the transactions with the card from last month."

The chatbot uses a Context-Free Grammar (CFG) to parse banking queries.

a) Analyse the ambiguity present in the given sentence using CFG

A CFG represents a sentence using grammatical production rules such as:

S → VP

VP → V NP

NP → Det N

NP → NP PP

PP → P NP

The sentence can be represented as:

Show [me] [the transactions] [with the card] [from last month]

The major ambiguity comes from the prepositional phrases (PPs):

- with the card
- from last month

Possible Interpretation 1

```
transactions
  |
  +--- with the card
  |
  +--- from last month
```

Meaning:

Show transactions that were made with the card and occurred last month.

This is the most likely banking interpretation.

Possible Interpretation 2

The phrase "with the card" could be interpreted as modifying the action show, meaning:

Show the transactions using/along with the card.

Possible Interpretation 3

The phrase "from last month" could potentially attach to the card-related noun phrase rather than directly to transactions.

This produces multiple possible parse trees because CFG primarily represents syntactic structure and does not inherently determine which interpretation is semantically most appropriate.

Main ambiguity

The ambiguity is therefore primarily caused by prepositional phrase attachment.

b) Evaluate the limitations of CFG in handling real-world conversational queries

A basic CFG has several limitations in a banking chatbot.

1. Structural ambiguity

CFG may generate multiple valid parse trees for the same sentence.

For example:

transactions with the card

can have different attachment structures.

2. No probability

A basic CFG treats valid grammatical structures without naturally assigning probabilities to them.

It cannot easily determine:

Which interpretation is most likely?

3. Subject–verb agreement

Basic CFG rules such as:

S → NP VP

do not automatically enforce agreement.

For example:

The customer check balance.

could require agreement information to reject the incorrect form.

4. Conversational language

Real users may say:

Show transactions last month

or:

Can you show me what I spent last month?

A rigid CFG may not adequately handle variations, fragments, ellipsis, or informal expressions.

5. Long queries

For long sentences, the number of possible parse structures can increase considerably, making parsing computationally expensive.

6. Lack of semantic knowledge

CFG cannot by itself determine that:

transactions + last month

represents a banking transaction query involving a time constraint.

c) Evaluate and propose an improved solution using PCFG, Feature Structures, or Earley Parsing: A better solution is to combine:

- PCFG
- Feature Structures
- Earley Parsing

rather than relying on basic CFG alone.

1. PCFG for ambiguity resolution

A Probabilistic Context-Free Grammar assigns probabilities to grammar rules.
For example:
VP → V NP PP 0.60
VP → V NP 0.30
NP → NP PP 0.10
When multiple parse trees are possible, the system can select the parse with the highest probability.
For the banking query, the model can learn that:
transactions + from last month
is a highly probable interpretation.

2. Feature Structures for agreement

Feature structures can store grammatical properties such as:
Number = singular/plural
Person = first/second/third
Tense = past/present/future
For example:
NP:
Number = plural
must agree with:
VP:
Number = plural
This allows the parser to reject structures with incorrect subject–verb agreement.

3. Earley Parsing

Earley parsing is useful because it can efficiently handle general CFGs and can represent multiple possible parses.
It is especially useful when the input is:

- Ambiguous
- Long
- Partially complete
- Conversational

The parser maintains partial states rather than repeatedly constructing complete parses from scratch.

Proposed Architecture

User Query
↓
Text Preprocessing
↓
Feature Extraction
↓
Earley Parser + CFG
↓
PCFG Probability Ranking
↓
Feature Agreement Checking
↓
Semantic Interpretation
↓
Banking Intent
↓
Transaction Search
↓
Response

d) Justify how the proposed method improves accuracy and efficiency in an industry setting

The combined approach improves the banking chatbot in several ways.

Accuracy

PCFG selects the most probable interpretation among ambiguous structures.

Feature structures improve grammatical consistency.

Earley parsing handles complex and ambiguous structures more systematically.

Efficiency:

Earley parsing uses dynamic programming to avoid unnecessarily repeating parsing work.

Better banking interpretation

For:

"Show me the transactions with the card from last month."

the system can produce a structured representation such as:

INTENT = TRANSACTION_QUERY
OBJECT = TRANSACTIONS
INSTRUMENT = CARD
TIME = LAST_MONTH
CUSTOMER = USER

This representation can then be passed to the banking transaction service.

Industry benefits

- More accurate customer responses
- Better intent detection
- Reduced incorrect transactions
- Better handling of natural language
- Improved scalability
- Better user experience

Conclusion:

A combination of CFG + PCFG + Feature Structures + Earley Parsing provides a stronger solution than a basic CFG-only chatbot.

2. Earley Parsing for Voice Assistant Systems

Given Command

"Book a flight to Delhi with a window seat."

a) Analyse how the given sentence can lead to multiple parse structures

The sentence contains the following major components:

Book
↓
a flight
↓
to Delhi
↓
with a window seat

The main ambiguity is caused by the prepositional phrase:

"With a window seat"

Interpretation 1 — Flight requirement

The most natural interpretation is:

Book a flight to Delhi and request a window seat.

Structure:

```
VP
├── Verb: Book
├── NP: a flight
│   └── PP: to Delhi
└── PP: with a window seat
```

Interpretation 2 — Flight description

The phrase:

with a window seat could theoretically be attached to the noun phrase flight:

NP
├── a flight
└── PP: with a window seat
Although both structures can be syntactically possible, contextual knowledge strongly Favors the interpretation that window seat is a booking preference.

b) Analyse the limitations of top-down parsing in this real-time application
Top-down parsing starts from the start symbol and attempts to generate the input.
For example:
S
↓
VP
↓
V NP
↓
...
Limitation 1: Backtracking
If the parser chooses the wrong production rule, it may have to backtrack and try another rule.This increases processing time.
Limitation 2: Ambiguity
Several possible structures may be explored before finding the correct one.
Limitation 3: Partial input
Voice assistants receive speech incrementally.
For example:
Book a flight...
followed later by:
to Delhi...
A traditional top-down parser may not efficiently handle such incomplete input.
Limitation 4: Real-time constraints
Voice assistants need rapid responses. Excessive backtracking can increase latency.
Limitation 5: Long commands
As command length and grammatical complexity increase, unnecessary search can increase.

c) Analyse how Earley parsing can improve handling of ambiguity and partial inputs
Earley parsing is a dynamic-programming parsing technique designed for general CFGs.
It maintains states containing information about:

- What grammar rule is being processed
- What part has already been recognized
- What input position is currently being processed

Three major operations

1. Predictor

Predicts possible grammar rules.

2. Scanner

Matches grammar symbols against the input.

3. Completer

Completes previously predicted structures.

This allows the parser to maintain multiple possible interpretations without repeatedly restarting the entire parse.

Handling partial input

Suppose the user says:

"Book a flight…"

The parser can maintain a partial state while waiting for additional speech.

After:

"To Delhi…"

the parser extends the existing analysis.

Finally:

"With a window seat."

adds the preference.

This makes Earley parsing useful for conversational systems.

d) Compare top-down parsing vs Earley parsing

Feature	Top-Down Parsing	Earley Parsing
Parsing direction	Starts from grammar start symbol	Processes input while maintaining states
Ambiguity	Can require extensive backtracking	Can maintain multiple parses
Partial input	Less suitable	Better suited
Backtracking	Can be significant	Reduced through dynamic programming
General CFG support	Depends on implementation	Yes
Real-time conversational input	Less suitable	More suitable
Long/complex sentences	Can become inefficient	More systematic
Industry suitability	Simple command grammars	Complex conversational grammars

Conclusion:

For a voice assistant handling ambiguous and incomplete spoken commands, Earley parsing provides a more flexible approach than basic top-down parsing, particularly when the grammar is general and the input arrives incrementally.

3. Healthcare NLP System Using CFG, PCFG and Feature Structures

Given Medical Sentence

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The system must extract:

- Diagnosis
- Treatment/action
- Location
- Relationships between entities
- Agreement information
- Real-time information

a) Design a complete NLP architecture

A suitable architecture is:

Medical Report

↓

Text Preprocessing

↓

Sentence Segmentation

↓

Tokenization

↓

Medical Named Entity Recognition

↓

CFG Parsing

↓

Feature Structure Validation

↓

PCFG Ambiguity Resolution

↓

Efficient Parsing / Earley Parsing

↓

Sub-Categorization Analysis

↓

Semantic Role / Relation Extraction

↓

Structured Clinical Information

↓

Hospital Database / Decision Support

1. CFG for syntactic structure

CFG provides the basic syntactic structure.

For example:

S → NP VP

NP → Det N

NP → NP RelClause

VP → V NP

VP → V GerundPhrase

The sentence can be divided into:

Subject:

The doctor who reviewed the patient last week

Verb:

recommends

Actions:

starting medication

scheduling a follow-up visit in Chennai

2. PCFG for ambiguity resolution

The sentence contains multiple possible attachment relationships.

For example:

last week

is associated with the review event.

Similarly:

in Chennai

is associated with the follow-up visit.

PCFG assigns probabilities to competing syntactic structures and selects the most likely interpretation.

3. Feature Structures for agreement

Feature structures can represent:

Person

Number

Gender

Tense

For example:
Doctor:
Number = Singular
Person = Third
Therefore:
recommends
is consistent with a third-person singular subject.
The feature structure helps prevent incorrect analyses such as:
The doctor recommend.
when the intended grammar requires:
The doctor recommends...

4. Efficient parsing
For long medical sentences, an efficient parser such as Earley parsing can be integrated.
This is useful for:

- Long sentences
- Ambiguous structures
- Complex subordinate clauses
- Partial medical text

b) Step-by-step workflow for the given sentence
Step 1: Input
The doctor who reviewed the patient last week recommends
starting medication and scheduling a follow-up visit in Chennai.

Step 2: Tokenization
The system identifies tokens:
The
doctor
who
reviewed
the
patient
last
week
recommends
starting
medication
and
scheduling

a
follow-up
visit
in
Chennai

Step 3: Named Entity Recognition
The system identifies important entities:
Doctor → Medical/Person Entity
Patient → Medical/Person Entity
Medication → Treatment Entity
Chennai → Location

Step 4: CFG Parsing
The parser identifies the major structure:
NP → The doctor + relative clause
VP → recommends + coordinated actions
The relative clause:
who reviewed the patient last week
modifies:
doctor

Step 5: Feature Checking
The subject is:
doctor
Number = Singular
Person = Third
The main verb is:
recommends
which agrees with the singular third-person subject.

Step 6: PCFG Disambiguation
The PCFG ranks competing interpretations.
It identifies:
last week → reviewed
in Chennai → follow-up visit
as the most likely contextual relationships.

Step 7: Sub-Categorization

The system analyzes the verb:
recommend
A useful frame is approximately:
recommend
Agent/Experiencer → Medical professional
Action/Theme → Treatment or medical action
For:
recommends starting medication
the doctor is associated with the recommendation and:
starting medication
is the recommended action.

Step 8: Semantic Representation

The system can generate:
AGENT = Doctor
ACTION = Start Medication
ACTION = Schedule Follow-up Visit
LOCATION = Chennai
TIME = Last Week
PATIENT = Patient
A structured representation could be:

```
RECOMMEND (
  Agent = Doctor,
  Actions = [
    START(Medication),
    SCHEDULE (FollowUpVisit, Chennai)
  ]
)
```

Step 9: Final Structured Output

Information	Extracted Value
Agent	Doctor
Related Patient	Patient
Treatment	Starting medication
Follow-up Action	Schedule follow-up visit
Location	Chennai
Time Reference	Last week
Main Intent	Medical recommendation

Important observation

The sentence does not explicitly provide a diagnosis.

Therefore, the system should not invent a diagnosis. The diagnosis field should be:

Diagnosis = Not explicitly stated

This is important in a healthcare application.

c) Method for sub-categorization, real-time processing, accuracy and scalability

1. Incorporating sub-categorization frames

Medical verbs should be associated with expected arguments.

Example: recommend

```
Recommend (
   Agent,
   Treatment/Action
)
```

Example:

Doctor recommends medication.

Representation:

Agent = Doctor

Action = Medication

Example: prescribe

```
Prescribe (
   Doctor,
   Medicine,
   Patient
)
```

Example: schedule

```
Schedule (
   MedicalStaff,
   Appointment
)
```

These frames help the system determine the relationship between medical actions and entities.

2. Real-time processing

A hospital system requires low latency.

A suitable architecture is:

Hospital Input

↓

Streaming Preprocessing

↓

Fast Tokenization

↓

NER + Parsing

↓

Semantic Extraction

↓

Clinical Database

Techniques include:

- Incremental parsing
- Efficient Earley-style parsing
- Caching
- Parallel processing
- Batch processing where appropriate
- Pre-trained medical NLP models

3. Accuracy improvement

Accuracy can be improved through:

Domain-specific grammar

Create grammar rules specifically for medical language.

PCFG

Use probabilities to select the most likely interpretation.

Feature Structures

Enforce grammatical agreement.

Medical NER

Recognize:

- Medicines
- Diseases
- Symptoms
- Procedures
- Locations
- Healthcare professionals

Medical ontology

Map extracted concepts to standardized medical concepts.

Confidence scores

If the system is uncertain, flag the result for human review.

4. Scalability in a hospital environment

A scalable system can use:

```
                    Hospital Data
                          ↓
               NLP Processing Layer
        ┌─────────────────┴─────────────────┐
        ↓                                   ↓
  Real-Time Queue                   Batch Processing
        ↓                                   ↓
    NLP Engine                         NLP Engine
        └─────────────────┬─────────────────┘
                          ↓
                   Structured Data
                          ↓
                  Clinical Database
                          ↓
               Decision Support System
```

The system can distribute NLP processing across multiple services so that many medical reports can be processed simultaneously.

Final Comparison of the Three Solutions

Requirement	**Basic CFG**	**Improved Approach**
Syntactic structure	✓	✓
Ambiguity handling	Limited	PCFG
Agreement handling	Limited	Feature Structures
Long sentences	Can be inefficient	Earley/Dynamic Parsing
Medical relationships	Limited	Sub-categorization
Real-time processing	Limited	Incremental/Efficient Parsing
Domain knowledge	Limited	Medical ontology
Scalability	Moderate	Distributed NLP architecture